

Human Brain Ancestral Barcodes

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eLife Assessment

This study presents a **valuable** conceptual approach that cell lineage can be determined using methylation data. However, the evidence supporting the claims of the author remains **incomplete** after revision. If clarified further as described in the reviews, this approach could be of broad interest to neuroscientists and developmental biologists.

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Abstract

Dynamic CpG methylation “barcodes” were read from 15,000 to 21,000 single cells from three human male brains. To overcome sparse sequencing coverage, the barcode had ~31,000 rapidly fluctuating X-chromosome CpG sites (fCpGs), with at least 500 covered sites per cell and at least 30 common sites between cell pairs (average of ~48). Barcodes appear to start methylated and record mitotic ages because excitatory neurons and glial cells that emerge later in development were less methylated. Barcodes are different between most cells, with average pairwise differences (PWDs) of ~0.5 between cells. About 10 cell pairs per million were more closely related with PWDs < 0.05. Barcodes appear to record ancestry and reconstruct trees where more related cells had similar phenotypes, albeit some pairs had phenotypic differences. Inhibitory neurons showed more evidence of tangential migration than excitatory neurons, with related cells in different cortical regions. fCpG barcodes become polymorphic during development and can distinguish between thousands of human cells.

Introduction

Cell lineages outline tissue development. Complete fate maps are possible by direct observation for small organisms such as *C. elegans*, but various elegant experimental fate markers are employed for larger tissues and longer time intervals (1 ). For human tissues, prior experimental manipulations are impractical, and genomic alterations are employed. Somatic mutations mark subclones and their fates can be reconstructed with DNA sequencing. Recent advances in single cell technologies potentially allow fate map reconstruction at single cell resolution.

Here we show how fluctuating CpG (fCpG) DNA methylation (2 [↗](#)) could be used as dynamic barcodes to study human brain development using single cell epigenomes annotated with their locations and phenotypes (3 [↗](#)). DNA methylation patterns are usually copied between cell divisions, but replication errors are much higher compared to base replication, allowing for more differences between daughter cells. DNA methylation modulates expression and their patterns can be used to infer cell phenotypes (3 [↗](#), 4 [↗](#)), but most fCpG sites are present outside of genes or in unexpressed genes. Criteria for our fCpG barcode are as follows: 1) a defined initial pattern in a progenitor cell; 2) polymorphic changes upon cell division; 3) adequate polymorphism to distinguish between most cells; and 4) capability to record ancestry.

The brain has several features that facilitate barcode development and validation. Foremost, there is extensive single cell methylation data, with thousands of cells annotated by locations and phenotypes (3 [↗](#)). Although billions of cells are present in an adult brain, lineage trees are compact because growth is largely neonatal. The brain also allows for serial “stopwatch” barcode sampling because development roughly follows a caudal to rostral pattern, and groups of neurons characteristically stop dividing and differentiate at different times and locations (5 [↗](#)). Brainstem neurons emerge early (6 [↗](#)) and their barcodes should most resemble the initial progenitor state, whereas the stopwatch runs longer for excitatory neurons that appear later in development. To facilitate presentation, barcode performance is summarized as follows: The brain fCpG barcode initializes as predominately methylated in the progenitor cell and becomes polymorphic with more diverse barcodes in excitatory neurons that emerge later in development. The barcode becomes sufficiently polymorphic to uniquely distinguish between most sampled brain cells, and barcoded cells organize into lineage trees.

Results

fCpG barcode identification

Barcode development was limited by the sparse single cell data (3 [↗](#)), with < 5% of CpG sites sequenced, often with only a single read. Sparse coverage was mitigated with X-chromosome fCpG sites because only a single read can infer a binary (0,1) state in male individuals. Autosomal CpG sites require at least 2 reads to infer three possible states (0, 0.5, 1). The X-chromosome also simplifies the identification of polymorphic fCpGs because many neurons have different binary states if average methylation is between 0.25 and 0.75 in bulk WGBS adult male neurons reference data (4 [↗](#)).

CpG sites (N~116,000), with average methylation between 0.25 and 0.75 in bulk neurons from seven males (4 [↗](#)), were further filtered by discarding more stable CpG sites with average methylation less than 0.2 or more than 0.8 for all cells, inhibitory neurons, and excitatory neurons in brain H02. The ~79,000 CpGs were further filtered to remove sites with average methylation less than 0.3 or greater than 0.7 in brain H01, and ~31,000 fCpG sites were used for analysis.

fCpG site methylation appears neutral because they are predominately intergenic, with 16% within genes or promoters (File S1). Epigenomes from 15,434 to 21,836 cells were downloaded from three male brains with a general criterion of allc.tsv.gz file sizes 90 mb or larger (Table 1). Neurons were preferentially sampled whereas glial cells were sometimes excluded (File S2). Analyzed cells had at least 500 fCpGs (average ~1,100), with pairwise distances (PWDs) calculated between cell pairs when at least 30 fCpGs were comparable (average ~48 fCpGs per cell pair). Each cell, annotated by its provided phenotype and location, is characterized by its fCpG methylation level and its PWDs from other cells. A PWD of 0 is a perfect match and 0.5 indicates randomization. fCpG methylation was variable between cells with averages of ~58% for all three brains (Fig 1A [↗](#)). The 73 to 197 million possible cell pair comparisons revealed polymorphic barcodes with average PWDs of ~0.47 between cells (Fig 1B [↗](#)).

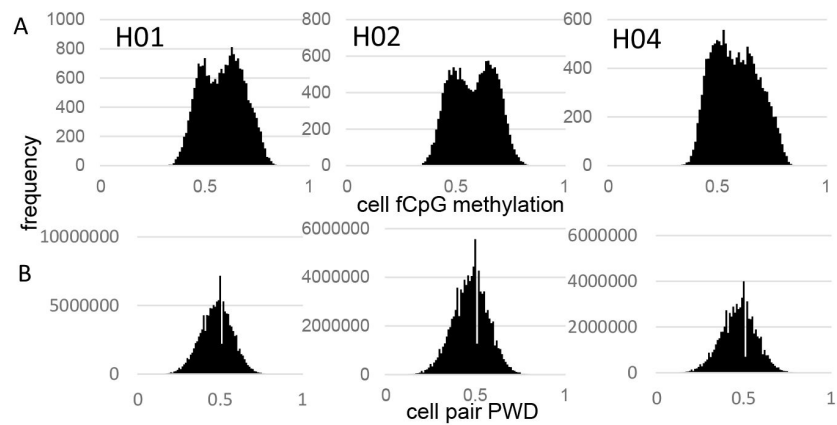


Fig 1

fCpG barcode methylation

- A)** Barcode methylation was variable between cells with averages $\sim 50\%$
- B)** Most cells had different barcodes with average PWDs ~ 0.5 .

fCpG barcodes initialize methylated and change with cell division

fCpG barcode patterns were similar between the brains, and data are presented for H01, with H02 and H04 presented in [Figs S1](#) and [S2](#). Methylation was variable between cells of the same type and average methylation was highest in the pons (PN) and thalamus (THM, intermediate for other inhibitory neurons, and lowest for excitatory neurons, glial cells, and cerebellar cells ([Fig 2A](#)). Outer layer cortical excitatory neurons (L2_3) that are made later during development were less methylated than inner cortical excitatory neurons (L4_6) that appear earlier. Predominately methylated individual fCpG sites were common in hindbrain neurons (PN, THM), less frequent in other inhibitory neurons, and rare in excitatory neurons ([Fig 2B](#)).

The methylation hierarchy is consistent with a barcode initialized with predominately methylated fCpGs in a brain progenitor cell. Barcodes become progressively demethylated and are fixed when their cells stop dividing and differentiate, which occurs at different times and places during brain development. Barcodes for each cell type had a range of methylation ([Fig 2A](#)), consistent with synchronous development rather than a strict stepwise process. Very simplistically, the hindbrain with mature neurons ([6](#)) forms early in development, followed by inhibitory neurons in the ganglionic eminences, and then excitatory neurons and glial cells in the cortex. Barcode methylation follows this temporal development and reconstruct when specific neuron types start to appear and reach their adult contents ([Fig 2C](#)). For example, after barcodes change from ~100 to ~70% methylated, most adult brainstem (PN) and inhibitory neurons are present but excitatory neurons are fewer, with very few adult outer layer (L2_3) neurons. Outer excitatory and glial progenitor cells are present ([7](#)), but their barcodes continue to demethylate until they stop dividing and differentiate later in development. This stopwatch like pattern, with more demethylated barcodes in later appearing cell types, was present in all three adult brains.

fCpG barcodes are polymorphic

A progenitor cell barcode is assumed to become increasingly polymorphic with subsequent divisions. This pattern was observed, with average barcode PWDs lowest in brainstem cells (PN), intermediate between other inhibitory neurons, and highest for excitatory neurons ([Fig 3A](#)). Most cells had different barcodes, with an overall average PWD of ~0.47 ([Fig 1B](#)). Cells of the same phenotype were more similar with lower average PWDs ([Fig 3A, 3B](#)), suggesting they are more related to each other and have common progenitors.

The human brain has billions of cells and relatively few cells were sampled from each region. Consistent with sparse sampling, cell pairs with nearly identical barcodes and smaller PWDs (< 0.05) were rare. To help distinguish between ancestry and chance, cells within and between brains were compared (Table 1). Closely related cell pairs were ~2.9 times more frequent within a brain (average ~9.8 per million) compared to between brains (average ~3.4 per million). Closely related cells had fewer matching fCpG sites (~35 compared to ~48 for all cell pairs) and were more common early in development when barcodes are more methylated ([Fig 3C](#)), indicating that lower barcode complexity favors matching. Overall, fCpG barcodes are sufficiently polymorphic to distinguish between most adult brain cells.

Brain lineage trees

It should be possible to reconstruct human brain development if barcodes record ancestry. fCpG barcodes from ~1,000 brain cells with different phenotypes yield trees with a standard phylogeny software that resemble caudal to rostral development ([Fig 4](#)). The trees are rooted by a progenitor with a fully methylated barcode, and branches progressively yield brainstem neurons (PN), a subset of excitatory lower (L4_6) neurons, thalamic neurons (THM), inhibitory neurons, cerebellar cells, and glial cells. Excitatory neurons branch last, and hippocampal neurons (CA, DG) that may divide postnatally ([8](#)) were at the terminus. Cells generally grouped by phenotype,

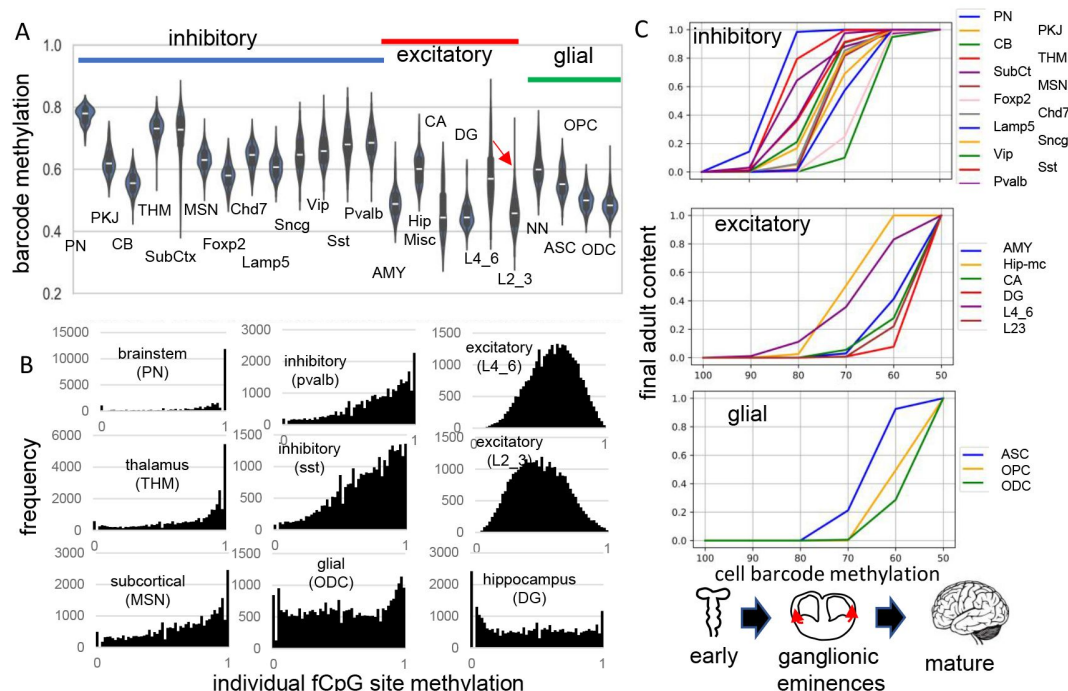


Fig 2

Higher fCpG barcode methylation in earlier emerging cells

A) Average barcode methylation was higher in the brainstem and inhibitory neurons. Barcode methylation was lower for excitatory neurons, cerebellar, and glial cells. Notably, average methylation was lower for outer cortical (L2_3) compared to earlier appearing inner (L4_6) cortical excitatory neurons. Abbreviations are as in reference 3, with L2_3 all outer and L4_6 all inner cortical excitatory neurons, and NN are non-neuronal cells other than ASC, OPC and ODC.

B) Most fCpGs appear to start methylated in a progenitor because nearly all individual fCpGs are methylated in inhibitory neurons in the subcortex (PN, THM, MSN). Many fCpGs in inhibitory neurons (pvalb, sst) are still predominately methylated. Few fCpGs in excitatory neurons that differentiate later in development are fully methylated. Glial cells that also emerge late in development, and hippocampal cells that may divide postnatally had variable methylation with both highly methylated and unmethylated fCpGs.

C) Barcodes are assumed to become fixed when their cells stop dividing and differentiate. Therefore, barcode methylation levels can indicate when neurons emerge during development, and can be correlated with a cartoon of physical caudal to rostral brain development. The X axis indicates the barcode methylation of individual cells and is assumed to roughly correlates with calendar time. The Y axis indicates the cumulative proportion of cells of each type present at each methylation level. A value of 0 indicates that cells of given type are not yet present and a value of 1 indicates the adult content of this cell type has been reached. At the start of development, inhibitory neurons (PN) in the pons with highly methylated barcodes appear first. More inhibitory neurons, made in the ganglionic eminences, appear and reach their final adult contents before many cortical excitatory neurons and glial cells appear. Notably, barcode methylation indicates many lower cortical layer neurons appear earlier in life relative to outer cortical neurons that reach adult levels late in development. Brain contents inferred by adult barcodes may differ from actual neonatal brains because neurons that die during development are not sampled in adult brains.

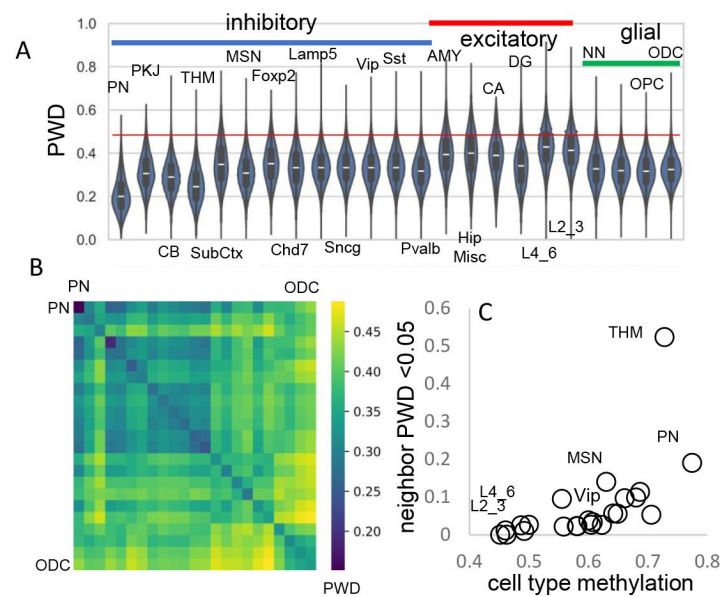


Figure 3

Related cell pairs

- A)** Most cells had different barcodes with average PWDs between cell pairs of ~ 0.5 . Cell pairs of the same phenotype had different barcodes but were on average more related to each other.
- B)** Heatmap showing that cells of the same phenotype are more related.
- C)** Cells that emerge early in development are more related and more methylated. Closely related nearest neighbors (PWD < 0.05) are numerically more common for more methylated cell types.

with some early appearing excitatory neurons admixed among inhibitory neurons. Similar trees were observed for H02 and H04, albeit with less separation between inhibitory and excitatory neurons for H04 (**Fig 4A**). Barcode lineage trees are largely consistent with expected sequential neuronal differentiation.

Cell lineage fidelity and cortical migration

Uncertain for mouse and human development is whether inhibitory and excitatory neurons originate from shared or distinct progenitors (3,9,10). Barcodes could potentially record neuronal differentiation patterns and lineage fidelity can be quantified by comparing most closely related cells or nearest neighbor cell pairs with PWDs < 0.05. The approach remains speculative due to several factors: the absence of direct experimental validation, limited experimental cell sampling, and the possibility that barcode similarities may arise by chance rather than reflecting true biological relationships. Lineage fidelity was high (>90%) for inhibitory neurons (**Fig 5A**). Excitatory lineage fidelity was slightly lower, indicating that some excitatory and inhibitory neurons may share common progenitors (10). Lineage trees indicate common progenitors are present earlier in development, and excitatory neurons that appear later do not have many closely related inhibitory neighbors (**Fig 4B**). The barcodes documented the known switching between inhibitory neuron subtypes (**Fig 5B**). Hindbrain, excitatory, and non-neuronal cells had more subtype lineage fidelity.

Barcodes can also potentially infer migration because their neurons are annotated by their adult locations. Daughter cells with similar barcodes could be sampled from the same region, or from different regions if migration occurred. Trees (**Fig 4C**) indicate that most neurons sampled from the brainstem and hippocampal regions are related and localized to their respective regions. Inhibitory neurons were scattered throughout the cortex, consistent with their differentiation in the ganglionic eminences and subsequent tangential migration to the cortex.

Nearest neighbor inhibitory cortical neuron pairs were found in the same cortical region ~25% of the time (**Fig 5C** and **5D**). Nearest neighbor excitatory neuron pairs were also scattered throughout the cortex, but less than inhibitory neurons, and were in the same cortical region ~50% of the time. The barcode data indicate more inhibitory rather than excitatory neuron tangential migration, but the extent of migration is uncertain due to sparse sampling and because barcodes can match by chance.

The poor ability to detect localized excitatory neuron radial cortical migration with ~1,000 cell whole brain trees (**Fig 4C**) may reflect that sparse sampling is unlikely to include multiple neurons from the same small clonal region that originated from common subventricular progenitors (radial unit hypothesis (11)). Greater localized excitatory neuron migration was seen when trees were reconstructed with more (~2,800) neurons, while inhibitory neurons still showed scattered tangential migration (**Fig 4D**). Neurons of the same subtype were still more related. Hence, lineage trees appear to increase their resolution with more cells, albeit related lower and upper excitatory neuron pairs were still uncommon, which may reflect the unlikely chance of sampling very small radial clonal units.

Discussion

fCpG barcodes are potential markers of somatic cell ancestry and not cell type classifiers, although cells of the same phenotype are often related because they originate from common progenitors (see new Supplement for fuller discussion). Dynamic barcodes would be useful to study human tissues, but testing their performance is difficult. Ideally, samples obtained at different times

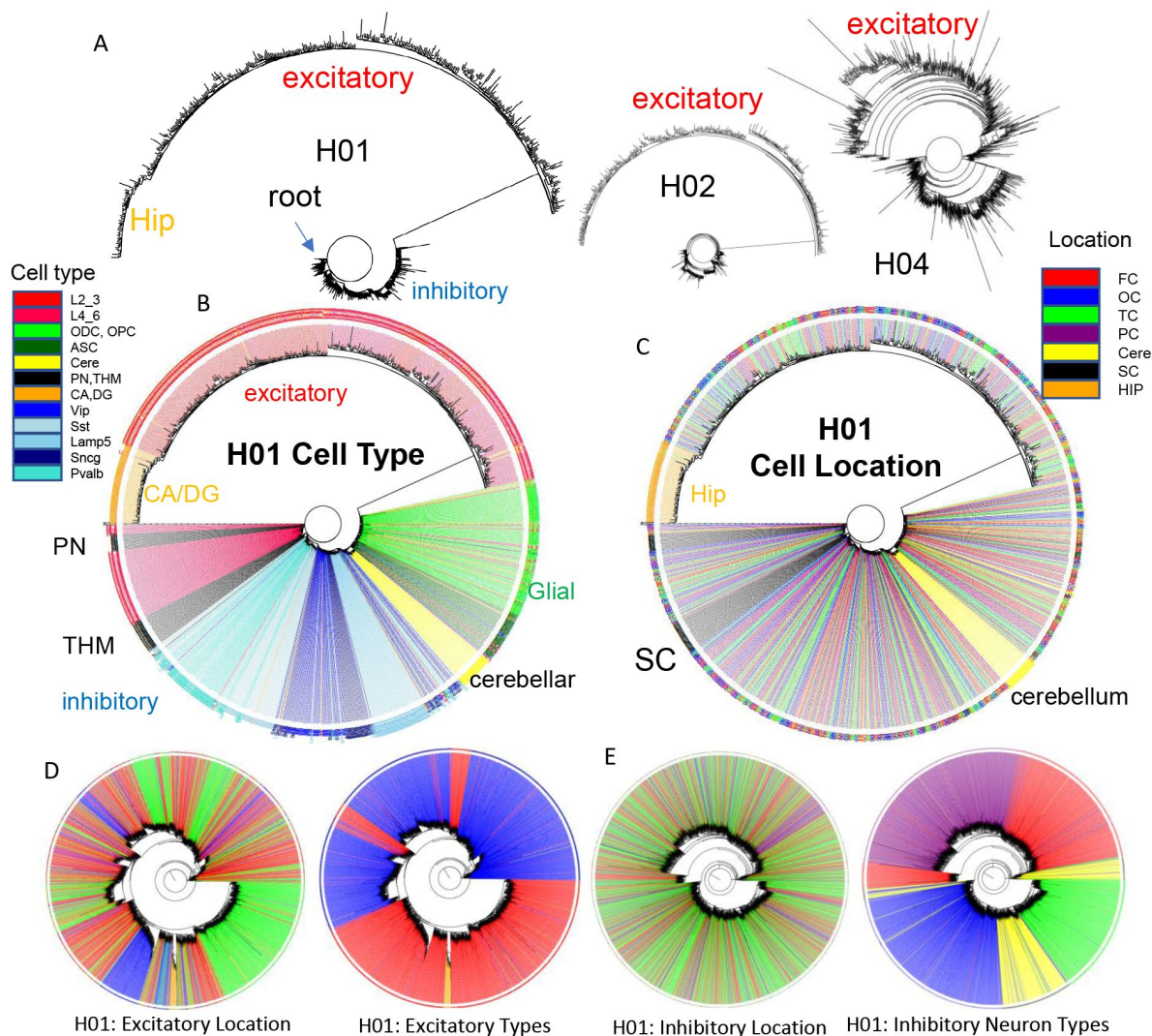


Figure 4

Brain trees

A) Barcodes from 960 cells form trees using IQtree (19) that are rooted by a fully methylated progenitor, and generally follow caudal to rostral brain development, with sequential branching of inhibitory neurons, cerebellar neurons, and excitatory neurons, with hippocampal neurons furthest from the start. Trees are similar between the brains, with H04 inferring less distance between inhibitory and excitatory lineages. The trees illustrate the ability to produce phylogenies with IQtree, but the phylogenies are limited by sparse cell sampling and that barcodes may be similar by chance. The degree of confidence was generally low, with bootstrap branch support typically less than 15%.

B) H01 tree with labeled cell types. Neuron types generally clustered by phenotypes with closely branching excitatory and inhibitory neurons more common earlier in development.

C) H01 tree with labeled cell locations. Related inhibitory and excitatory neurons can be found in different parts of the brain (FC = frontal (red), TC = temporal, OC = occipital (blue), PC = parietal, HIP = hippocampus (orange), cere = cerebellum (yellow), SC = subcortical (black)).

D) H01 tree with ~2,853 cortical excitatory neurons has more evidence of localized radial migration because related neurons are more often found in the same cortical region. Excitatory neurons cluster by subtype, and closely related lower and upper excitatory neurons were still few.

E) H01 tree with ~2,847 cortical inhibitory neurons still retains evidence of tangential migration with related neurons scattered throughout the cortex. Inhibitory neurons cluster by subtype with switching between some closely related pairs.

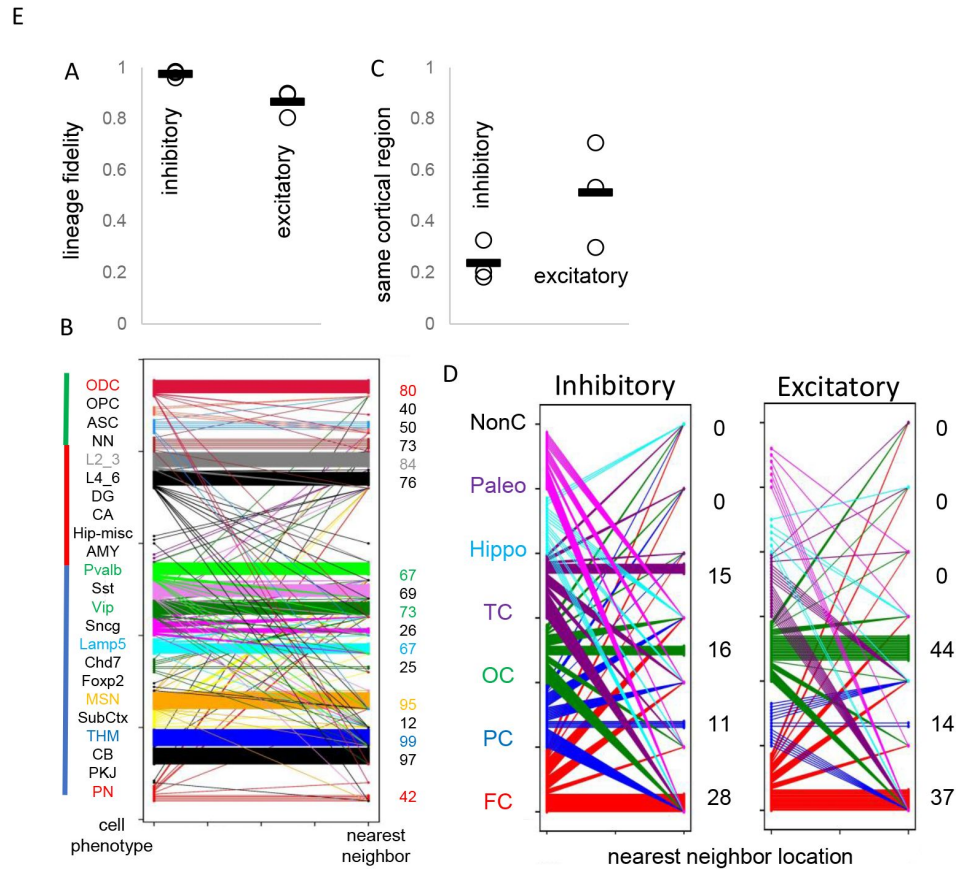


Figure 5

Lineage fidelity, migration, and differentiation

A) Inhibitory neurons have higher lineage fidelity because nearest neighbor pairs (PWD <0.05) were nearly always both inhibitory neurons. Excitatory neurons had slightly less lineage fidelity because a nearest neighbor was more often an inhibitory neuron. Data are for all three brains.

B) Nearest neighbor inhibitory neuron pairs often had subtype differences. More lineage fidelity was generally present for subcortical and excitatory neurons. Numbers indicate percent lineage subtype fidelity.

C) Nearest neighbor inhibitory and excitatory neuron pairs showed evidence of tangential migration because they were found in different cortical regions. The data indicate greater evidence of inhibitory neuron tangential migration than for excitatory neurons. However, the extent of migration is uncertain because of sparse sampling and because barcodes may be similar by chance. Data are for all three brains.

D) Nearest neighbor neurons were scattered in the cortex. Numbers indicate percent location fidelity. (NonC = non-cortical location, Paleo = paleocortex)

would document how they change. The brain facilitates barcode validation because it periodically stores neurons that stop recording at relatively defined times and locations (1,13). Specific neuron subsets recovered from the adult brain allow for sampling through time and before birth.

This serial sampling strategy facilitated fCpG barcode validation. The barcode appeared to start predominately methylated in multiple individuals and became sufficiently polymorphic to distinguish between thousands of neurons. Barcode changes appear to represent replication errors because they reconstruct lineage trees roughly consistent with caudal to rostral brain development. Barcode methylation may indicate when different neurons that survive to adulthood appear in the neonatal brain (Fig 2C).

The current barcode indicates that most inhibitory and excitatory neurons have relative distinct progenitors, consistent with the lineage dendrograms reconstructed with neuron specific methylation of the same data (3). There was also evidence for common inhibitory and excitatory progenitors (10), primarily for earlier emerging excitatory neurons. Tangential migration was also detected, manifested by inhibitory neurons with closely related barcodes in different cortical regions. Tangential excitatory neuron migration was also detected, albeit related excitatory neurons were more localized than inhibitory neurons. Tangential migration is also seen with sequencing studies that find neurons with specific mutations in multiple brain regions (14-17).

fCpGs more efficiently distinguish between cells than mutations due to higher replication error rates. Although average methylation decreases with time, both demethylation and remethylation are likely because fully demethylated neurons were not observed, and balanced fluctuating methylation is inferred in other tissues when CpG sites are ~50% methylated in bulk tissues (2). More adult divisions in brain cancers did not saturate the barcode with average fCpG methylation ~50% (Fig S3). Fluctuating methylation complicates lineage tracing but backmutations can be modeled for ancestral reconstructions. Lineage resolution could be improved by combining mutations and fCpGs.

Weaknesses of this study including very sparse cell sampling and lack of uniform CpG sites comparisons between neurons. Like many human fate markers studies, it is difficult to independently verify accuracy. Interpretation of the barcodes relies on several untested assumptions including relatively constant error rates between fCpG sites and through aging, neutrality, and a predominately fully methylated start in the progenitor cell. Epigenetic remodeling occurs after progenitors stop dividing (12), which could erase ancestral barcode information. However, neurons of the same type were both closely and remotely related, indicating that such epigenetic remodeling does not systematically alter the fCpG sites. In addition, fCpG barcodes appear to be relatively stable through aging (new Supplement). Inferred lineage trees (Fig 4) had relatively low statistical support for their branches and are presented to demonstrate that the barcodes are readily organized into trees with a commonly used phylogeny software.

Technical improvements such as targeted bisulfite sequencing of a limited number of informative fCpGs could lead to more consistent coverage and less expensive sequencing of more neurons. Single cell measurements of small numbers of fCpGs, and snMCode cell type specific CpG sites (3), could efficiently reconstruct human brain lineages, although back changes add complexity. A barcode of 100 fCpGs has enough complexity (2^{100} or $\sim 1 \times 10^{30}$) to potentially distinguish between most excitatory neurons, with less resolution early in development when cells are inherently more related.

The analysis of more brains can verify that a fCpG barcode starts predominately methylated in most individuals. A common initialized state could facilitate standardized human fate maps and comparisons between individuals. Many polymorphisms linked to brain abnormalities such as

autism are in neuronal proliferation, migration, and maturation pathways (18 [↗](#)), and this preliminary survey indicates lineage heterogeneity between individuals (**Fig 4A** [↗](#)). fCpG barcodes have been applied to the intestines, endometrium, and blood (2 [↗](#)), and could be found for multiple other tissue types, helping to unravel human development and aging.

Methods

Brain Single Cells

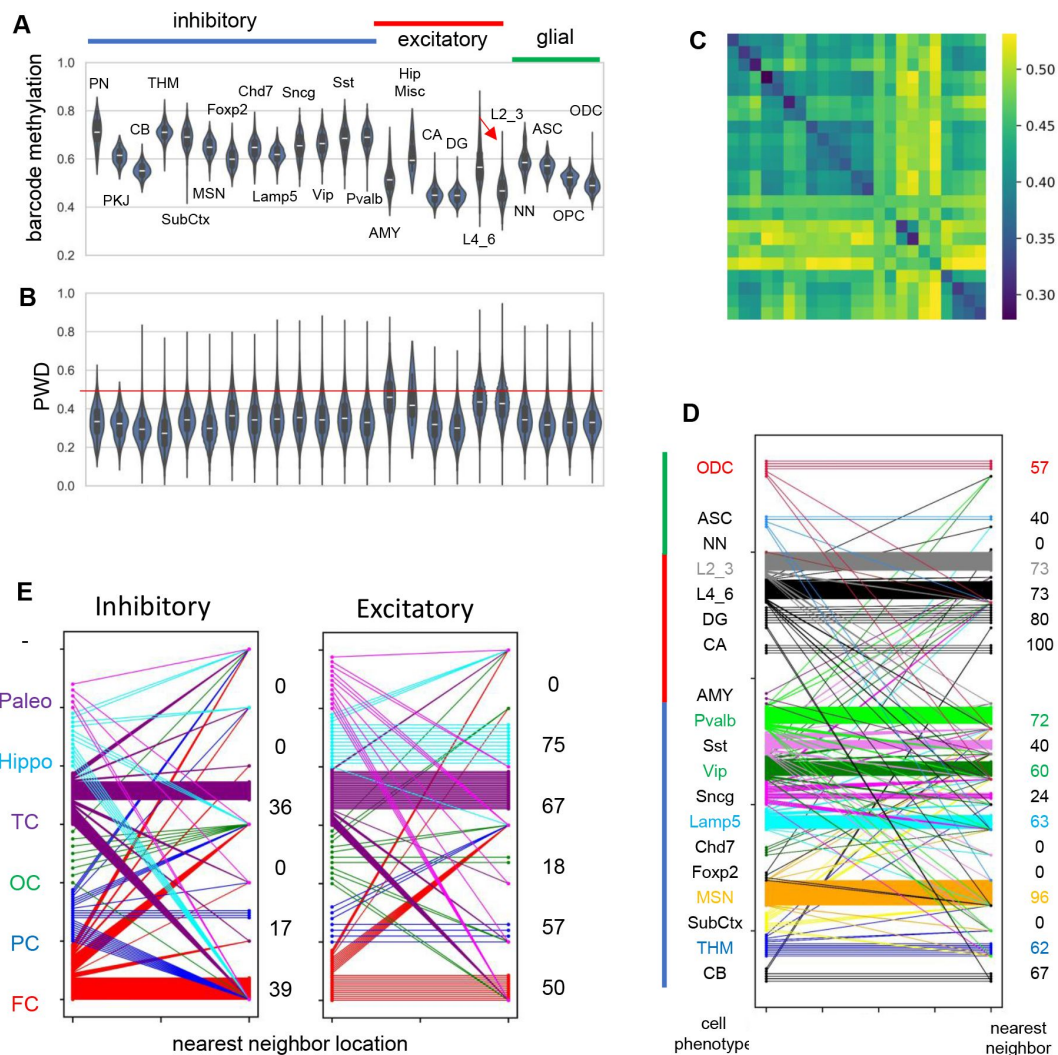
Single cells with their annotations and methylation at each fCpG site were read from single cell files downloaded from GEO (GSE215353) and supplemental files from reference 3. Lists of fCpG sites and data summarized for the Figures are in Supplemental File 2. The cells and methylation at the fCpG sites are in Supplemental Files 3-5. PWDs were calculated between all cell pairs with at least 30 matching fCpG sites, with PWD data matrices in Supplemental File 6. Additional details are provided in the new Supplement.

IQtree

IQtree (19 [↗](#)) tree was downloaded and run on a server with 64 cpus and 32 GB of memory. The model (GTR2+FO+G4) accounts for backmutation and binary data with missing values. Bootstraps were 1,000 per tree with 3,000 iterations for whole brain (~1,000 cells) trees and 1,000 iterations for inhibitory or excitatory (~2,800 neurons) trees. Trees (.treefile) were displayed with FigTree (<http://tree.bio.ed.ac.uk/software/figtree/> [↗](#)) with truncation of long branches (generally fewer than 10) for display purposes. The cells used for the trees are in Supplemental File 7.

Acknowledgements

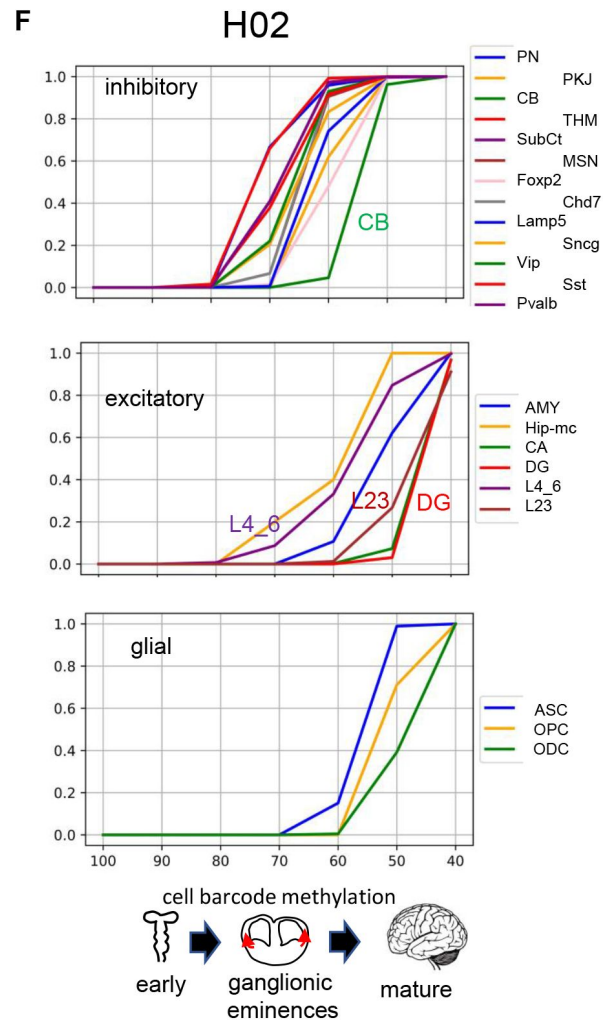
This work was supported by grants from the NIH (P01CA196569 and CA271237). I thank Drs. Trevor Graham and Heather Grant for useful discussions, and Omar Khan and Nikhil Krishnan for initial studies. The author thanks all of the researchers that helped produced the high quality very valuable and freely available data used for analysis.



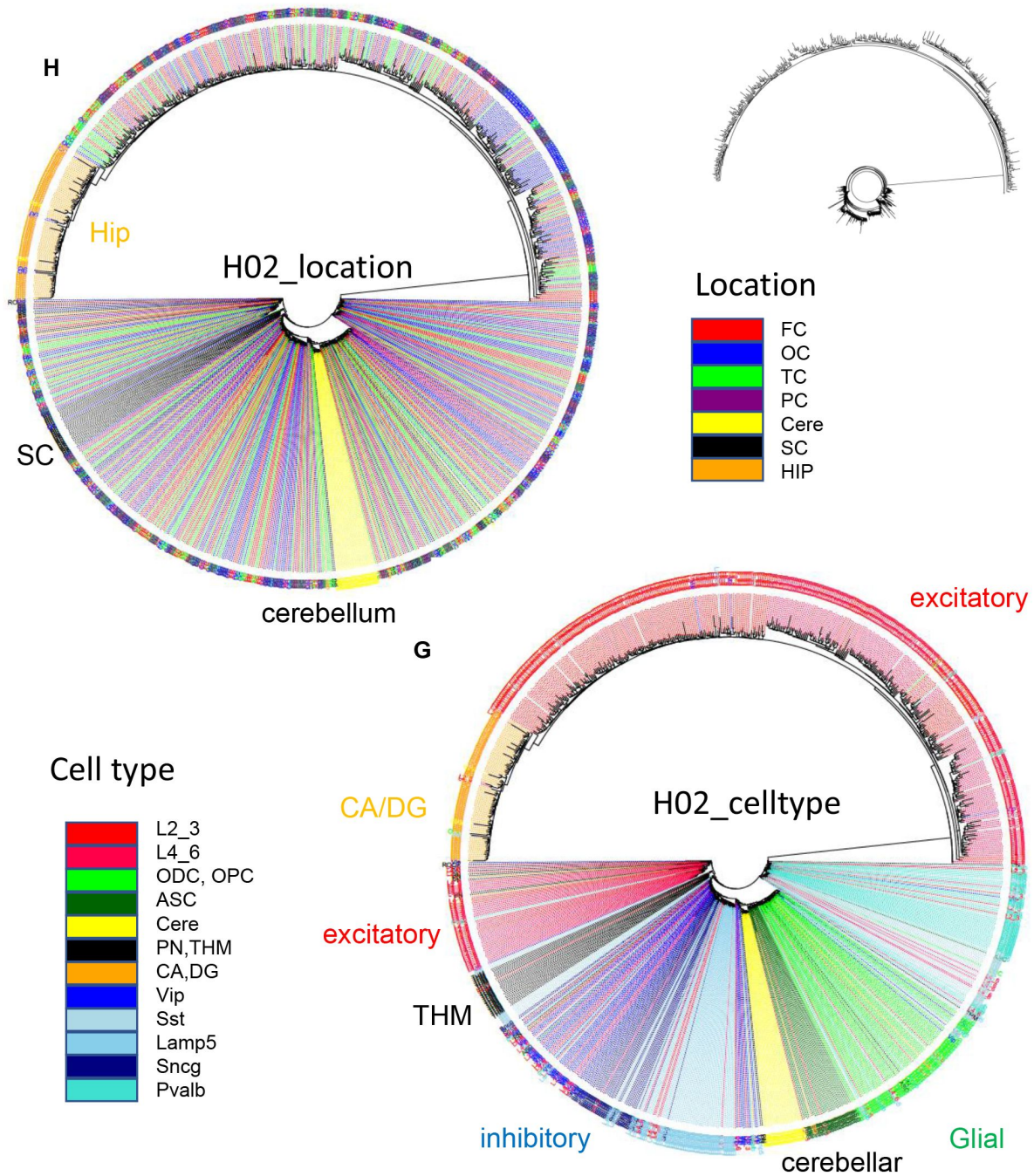
Supplementary Figure 1

H02 data

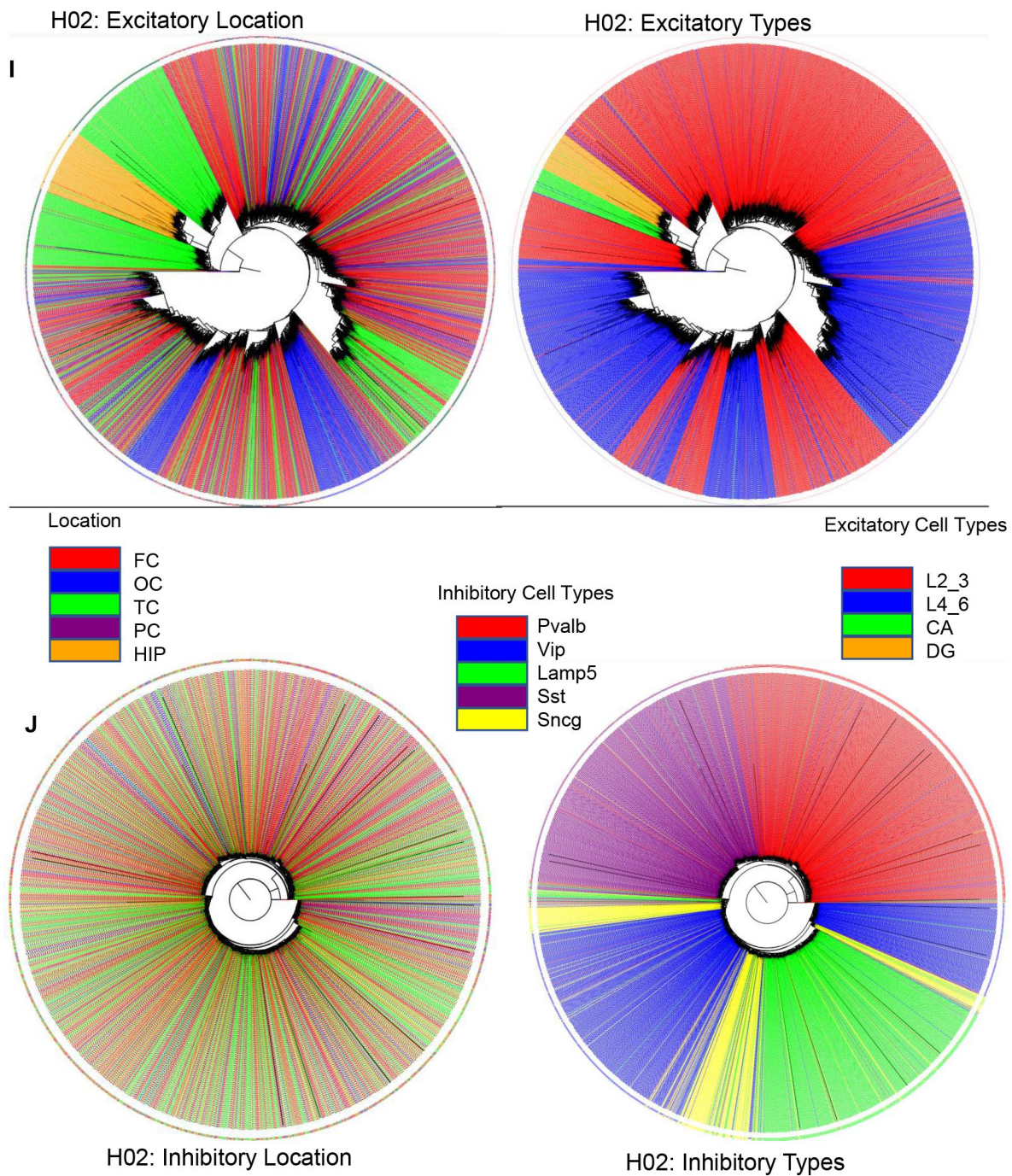
- A)** Barcode methylation for different cell types
- B)** PWDs between cells of the same type
- C)** PWDs between cell types
- D)** Lineage cell type fidelity between nearest neighbor pairs (PWD < 0.05)
- E)** Location fidelity between nearest neighbor pairs (PWD < 0.05)
- F)** Barcode methylation versus final adult brain content indicates that inhibitory neurons appear first and reach their adult levels before excitatory or glial cells.
- G)** Ancestral tree with 1,001 cells rooted at a fully methylated progenitor shows sequential branching with excitatory, then brain stem, inhibitory and cerebellar neurons, then glial cells, and finally excitatory neurons with hippocampal neurons at the end.
- H)** Related cells colocalize for brain stem, cerebellar, and hippocampal neurons. Inhibitory neurons are more scattered. Excitatory neurons are also scattered with some localization within the cortex (see I & J for trees with more neurons)
- I)** Ancestral tree with more (2,806) excitatory neurons shows more localization between related neurons in the hippocampus, occipital and temporal cortex. Related neurons also tend to have similar phenotypes.
- J)** Ancestral tree with more (2,788) inhibitory neurons still shows scattering between related neurons. Related neurons tend to have similar phenotypes.



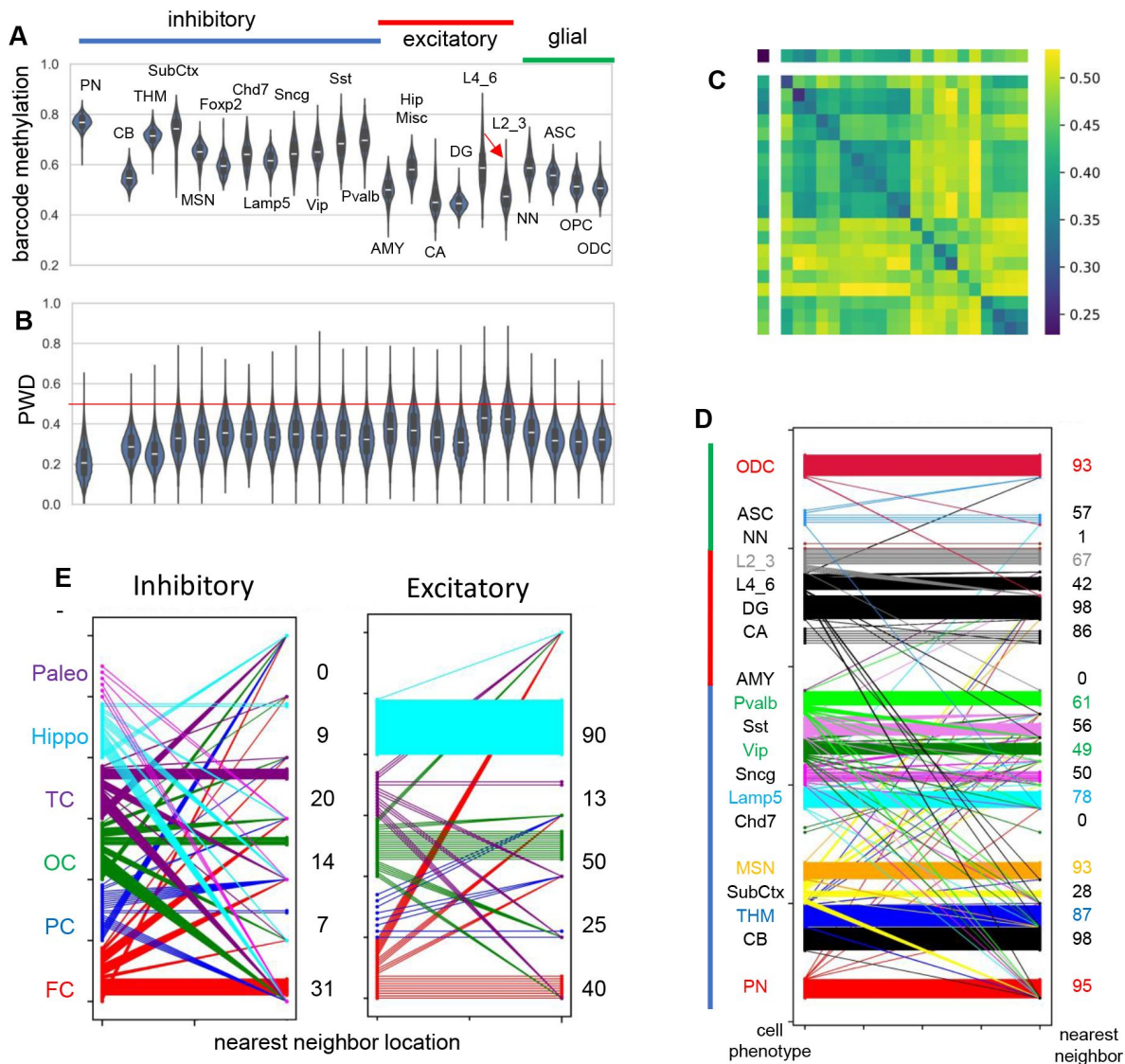
Supplementary Figure 1 (continued)



Supplementary Figure 1 (continued)



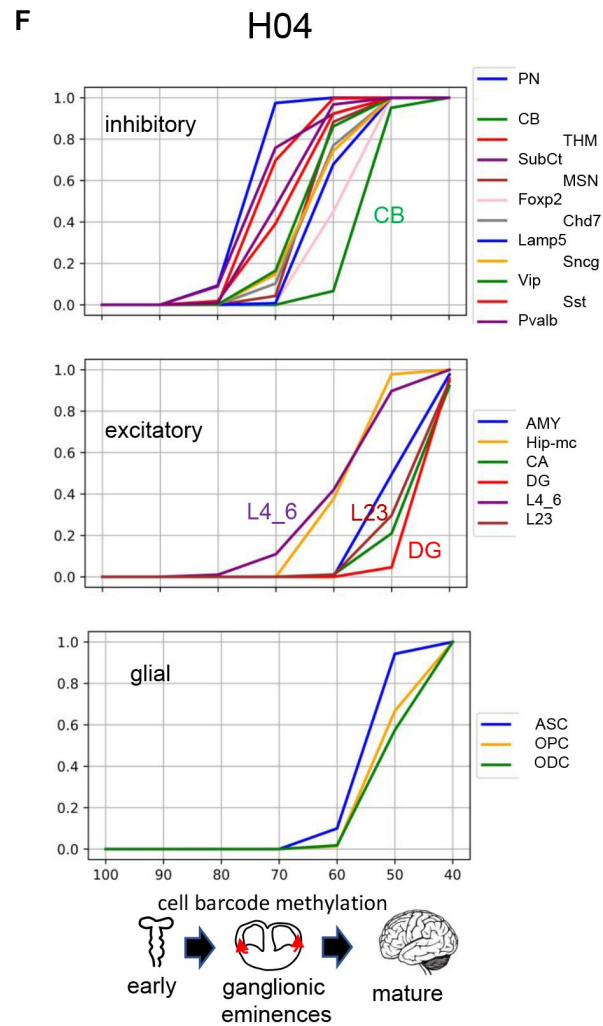
Supplementary Figure 1 (continued)



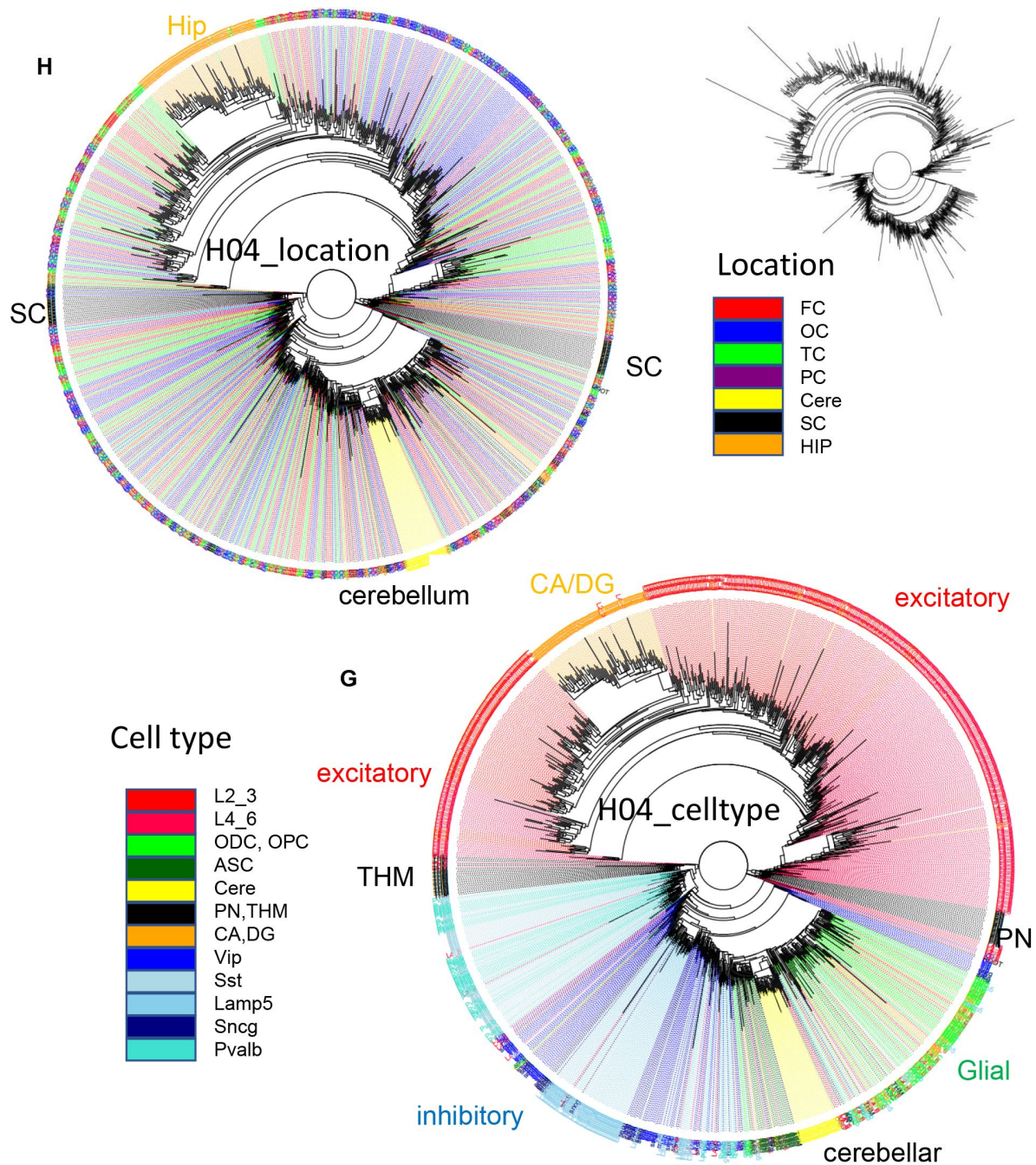
Supplementary Figure 2

H04 data

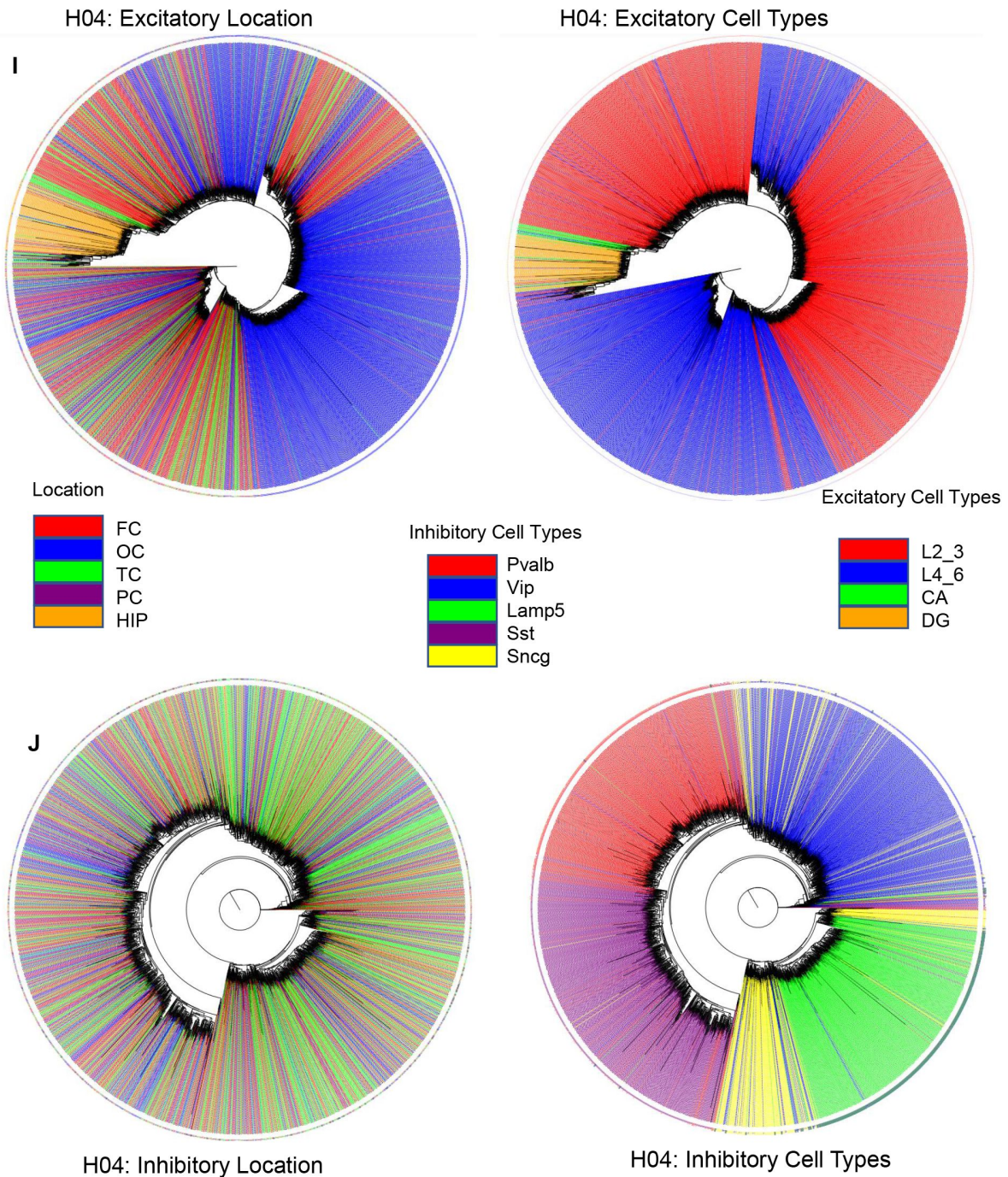
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Supplementary Figure 2 (continued)



Supplementary Figure 2 (continued)



Supplementary Figure 2 (continued)

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Reviewer #1 (Public review):

Summary:

In this manuscript, Shibata describes a method to assess rapidly fluctuating CpG sites (fCpGs) from single-cell methylation sequencing (sc-MeSeq) data. Assuming that fCpGs are largely consistent over time with changes induced by inheritable events during replication, the author infers lineage relationships in available brain-derived sc-MeSeq. Supplementing current lineage tracing through genomic and mitochondrial mosaic variants is an interesting concept that could supplement current work or allow additional lineage analysis in existing data.

However, the author failed to convincingly show the power of fCpG analysis to determine lineages in the human brain. While the correlation with cellular division and distinction of cell types appears plausible and strong, the application to detect specific lineages is less convincing. Aspects of this might be due to a lack of clarity in presentation and erroneous use of developmental concepts. However, without addressing these problems it is challenging for a reader to come to the same conclusions as the author.

On the flip side, this novel application of fCpGs will allow the re-use of existing sc-MeSeq to infer additional features that were previously unavailable, once the biological relevance has been further elucidated.

Strengths:

- Novel re-analysis application of methylation data to infer the status of fCpGs and the use as a lineage marker
- Application of this method to an innovative existing data set to benchmark this framework against existing developmental knowledge

Weaknesses:

- Inconsistent or erroneous use of neurodevelopmental concepts which hinders appropriate interpretation of the results.
- Somewhat confusing presentation at times which makes it hard to judge the value of this novel approach.

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Reviewer #3 (Public review):**Summary:**

Cell lineage tracing necessitates continuous visible tracking or permanent molecular markers that daughter cells inherit from their progenitors. To successfully trace cell lineages, it is essential to generate and detect sufficient new markers during each cell division. Thus, molecular cell lineages have been predominantly studied with stably inherited genetic markers in animal models and somatic DNA mutations in the human brain. DNA methylation is unstable across cell divisions and differentiation, and is hardly called barcodes. The use of "Human Brain Barcodes" in the title and across the whole paper lacks convincing evidence - it is questionable that CpG methylation is always stably inherited by daughter cells.

Strengths:

Analysis of DNA methylation.

Weaknesses:

The unstable nature of CpG methylation would introduce significant problems in inferring the true cell lineage. To establish DNA methylation as a means for lineage tracing, it is necessary to test whether the DNA methylation patterns can faithfully track cell lineages with in vitro differentiated & visibly tracked cell lineages.

The unreliable CpG methylation status also raises the question of what the "Barcodes" refer to in the title and across this study. Barcodes should be stable in principle and not dynamic across cell generations, as defined in the Reference #1. The CRISPR/Cas9 mutable barcodes or the somatic mutations may be considered barcodes, but the reviewer is not convinced that the "dynamic" CpG methylation fits the "barcodes" terminology. This problem is even more concerning in the last section of the results, where CpG status fluctuates in post-mitotic cells.

The manuscript frequently states assumptions in a tone of conclusions and interprets results without rejecting alternative hypotheses. For example, the title "Human Brain Barcodes" should be backed with solid supporting evidence. For another example, the author assumed that the early-formed brain stem would resemble progenitors better and have a higher average methylation level than the forebrain - however, this difference in DNA methylation status could well reflect cell-type-specific gene expression instead of cell lineage progression.

Other points:

(1) The conclusion that excitatory neurons undergo tangential migration is unclear - how far away did the author mean for the tangential direction? Lateral dispersion is known, but it is hard to believe that the excitatory neurons travel across different brain regions. More importantly, how would the author interpret shared or divergent methylation for the same cell type across different brain regions?

(2) The sparsity and resolution of the single-cell DNA methylation data. The methylation status is detected in only a small fraction ($\sim 500/31,000 = 1.6\%$) of fCpGs per cell, with only 48 common sites identified between cell pairs. Given that the human genome contains over 28 million CpG sites, it is important to evaluate whether these fCpGs are truly representative.

(3) While focusing on the X-chromosome may simplify the identification of polymorphic fCpGs, the confidence in determining its methylation status (0 or 1) is questionable when a CpG site is covered by only one read.

<https://doi.org/10.7554/eLife.101163.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife assessment:

Developing a reliable method to record ancestry and distinguish between human somatic cells presents significant challenges. I fully acknowledge that my current evidence supporting the claim of lineage tracing with fCpG barcodes is inadequate. I agree with Reviewer 1 that fCpG barcodes are essentially a cellular division clock that diverges over time. A division clock could potentially document when cells cease to divide during development, with immediate daughter cells likely exhibiting more similar barcodes than those that are less related. Although it remains uncertain whether the current fCpG barcodes capture useful biological information, refinement of this type of tool could complement other approaches that reconstruct human brain function, development, and aging.

Due to my lack of clarity, the fCpG barcode was perceived to be a new type of cell classifier. However, it is fundamentally different. fCpG sites are selected based on their differences between cells of the same type, while traditional cell classifiers focus on sites with consistent methylation patterns in cells of the same type. Despite these opposing criteria, fCpG barcodes and traditional cell classifiers may align because neuron subtypes often share common progenitors. As a result, cells of the same phenotype are also closely related by ancestry, and ex post facto, have similar fCpG barcodes. fCpG barcodes are complementary to cell type classifiers, and potentially provide insights into aspects such as mitotic ages, diversity within a clade, and migration of immediate daughters—information which is otherwise difficult to obtain. The title has been modified to “Human Brain Ancestral Barcodes” to better reflect the function of the fCpG barcodes. The manuscript is edited to correct errors, and a new Supplement is added to further explain fCpG barcode mechanics and present new supporting data.

Reviewer #1 (Public review):

I thank Reviewer 1 for his constructive comments. Major noted weaknesses were 1) insufficient clarity and brevity of the methodology, 2) inconsistent or erroneous use of neurodevelopmental concepts, and 3) lack of consideration for alternative explanations.

(1) The methodology is now outlined in detailed in a new Supplement, including simulations that indicate that the error rate consistent with the experimental data is about 0.01 changes in methylation per fCpG site per division.

(2) Conceptual and terminology errors noted by the Reviewers are corrected in the manuscript.

(3) I agree completely with the alternative explanation of Reviewer 1 that fCpGs are “a cellular division clock that diverges over ‘time’”. Differences between more traditional cell type classifiers and fCpG barcodes are more fully outlined in the new Supplement. Ancestry recorded by fCpGs and cell type classifiers are confounded because cells of the same phenotype typically have common progenitors—cells within a clade have similar fCpG barcodes because they are closely related. fCpG barcodes can compliment cell type classifiers with additional information such as mitotic ages, ancestry within a clade, and daughter cell migration.

Reviewer #1 (Recommendations for the authors):

(1) A lot of the interpretations suffer from an extremely loose/erroneous use of developmental concepts and a lack of transparency. For instance:

a) The thalamus is not part of the brain stem

Corrected.

b) The pons contains cells other than inhibitory neurons in the data; the same is true for the hippocampus which contains multiple cell types

Corrected to refer to the specific cell types in these regions.

c) The author talks about the rostral-caudal timing a lot which is not really discussed to this degree in the cited references. Thus, it is also unclear how interneurons fit in this model as they are distinguished by a ventral-dorsal difference from excitatory neurons. Also, it is unclear whether the timing is really as distinct as claimed. For instance, inhibitory neurons and excitatory neurons significantly overlap in their birth timing. Finally, conceptually, it does not make sense to go by developmental timing as the author proposes that it is the number of divisions that is relevant. While they are somewhat correlated there are potentially stark differences.

The manuscript attempts to describe what might be broadly expected when barcodes are sampled from different cell types and locations. As a proposed mitotic clock, the fCpG barcode methylation level could time when each neuron ceased division and differentiated. The wide ranges of fCpG barcode methylation of each cell type (Fig 2A) would be consistent with significant overlap between cell types. The manuscript is edited to emphasize overlapping rather than distinct sequential differentiation of the cell types.

d) Neocortical astrocytes and some oligodendrocytes share a lineage, whereas a subset of oligodendrocytes in the cortex shares an origin with interneurons. This could confound results but is never discussed.

The manuscript does not assess glial lineages in detail because neurons were preferentially included in the sampling whereas glial cells were non-systematically excluded. This sampling information is now included in the section “fCpG barcode identification”.

e) Neocortical interneurons should be more closely related in terms of lineage-to-excitatory neurons than other inhibitory neurons of, for instance, the pons. This is not clearly discussed and delineated.

This is not discussed. It may not be possible analyze these details with the current data. The ancestral tree reconstructions indicate that excitatory neurons that appear earlier in development (and are more methylated) are more often more closely related to inhibitory neurons.

f) While there is some spread of excitatory neurons tangentially, there is no tangential migration at the scale of interneurons as (somewhat) suggested/implied here.

The abstract and results have been modified to indicate greater inhibitory than excitatory neuron tangential migration, but that the extent of excitatory neuron tangential migration cannot be determined because of the sparse sampling and that barcodes may be similar by chance.

g) The nature of the NN cells is quite important as cells not derived from the neocortical anlage are unlikely to share a developmental origin (e.g., microglia, endothelial cells). This should be clarified and clearly stated.

The manuscript is modified to indicate that NN cells are microglial and endothelial cells. These cells have different developmental origins, and their data are present in Fig 2A, but are not further used for ancestral analysis.

(2) The presentation is often somewhat confusing to me and lacks detail. For instance:

a) The methods are extremely short and I was unable to find a reference for a full pipeline, so other researchers can replicate the work and learn how to use the pipeline.

The pipeline including python code is outlined in the new Supplement

b) Often numbers are given as ~XX when the actual number with some indication of confidence or spread would be more appropriate.

Data ranges are often indicated with the violin plots.

c) Many figure legends are exceedingly short and do not provide an appropriate level of detail.

Figure legends have been modified to include more detail

d) Not defining groups in the figure legends or a table is quite unacceptable to me. I do not think that referring to a prior publication (that does not consistently use these groups anyway) is sufficient.

The cell groups are based on the annotations provided with each single cell in the public databases.

e) The used data should be better defined and introduced (number of cells, different subtypes across areas, which cells were excluded; I assume the latter as pons and hippocampus are only mentioned for one type of neuronal cells, see also above).

The data used are present in Supplemental File 2 under the tab “cell summary H01, H02, H04”.

f) Why were different upper bounds used for filtering for H01 and H02, and H04 is not mentioned? Why are inhibitory and excitatory neurons specifically mentioned (Lines 61-66)?

The filtering is used to eliminate, as much as possible, cell type specific methylation, or CpG sites with skewed neuron methylation. The filtering eliminates CpG sites with high or low methylation within each of the three brains, and within the two major neuron subtypes. The goal is to enrich for CpG sites with polymorphic but not cell type specific methylation. This process is ad hoc as success criteria are currently uncertain. The extent of filtering is balanced by the need to retain sufficient numbers of fCpGs to allow comparisons between the neurons.

g) What 'progenitor' does the author refer to? The Zygote? If yes, can the methylation status be tested directly from a zygote? There is no single progenitor for these cells other than the zygote. Does the assumption hold true when taking this into account? See, for instance, PMID 33737485 for some estimation of lineage bottlenecks.

A brain progenitor cell can be defined as the common ancestor of all adult neurons, and is the first cell where each of its immediate daughter cell lineages yield adult neurons. The zygote is a progenitor cell to all adult cells, and barcode methylation at the start of conception, from the oocyte to the ICM, was analyzed in the new Supplement. The proposed brain progenitor cell with a fully methylated barcode was not yet evident even in the ICM.

(3) I am generally not convinced that the fCpGs represent anything but a molecular clock of cell divisions and that many of the similarities are a function of lower division numbers where the state might be more homogenous. This mainly derives from the issues cited above, the lack of convincing evidence to the contrary, and the sparsity of the assessed data.

Agree that the fCpG barcode is a mitotic clock that becomes polymorphic with divisions. As outlined in the new Supplement, ancestry and cell type are confounded because cells of the same type typically have a common progenitor.

a) There appears little consideration or modeling of what the ability to switch back does to the lineage reconstruction.

fCpG methylation flipping is further analyzed and discussed in the new Supplement.

b) None of the data convinced me that the observations cannot be explained by the aforementioned molecular clock and systematic methylation similarities of cell types due to their cell state.

See above

(4) Uncategorized minor issues:

a) The author should explain concepts like 'molecular clock hypothesis' (line 27) or 'radial unit hypothesis' (line 154), as they are somewhat complex and might not be intuitive to readers.

The molecular clock hypothesis is deleted and the radial unit hypothesis is explained in more detail in the manuscript.

b) Line 32: '[...] replication errors are much higher compared to base replication [...]'. I think this is central to the method and should be better explained and referenced. Maybe even through a schematic, as this is a central concept for the entire manuscript.

The fCpG barcode mechanics are better explained in the new Supplement. With simulations, the fCpG flip rate is about 0.01 per division per fCpG.

c) Line 41: 'neonatal'. Does the author mean to say prenatal? Most of the cells discussed are postmitotic before birth.

Corrected to prenatal.

d) Line 96: what does 'flip' mean in this context? Please also see the comment on Figure 2C.

Edited to “chage”

e) Lines 134-135: I am not sure whether the author claims to provide evidence for this question, and I would be careful with claims that this work does resolve the question here.

Have toned down claims as evidence for my analysis is currently inadequate.

f) Lines 192-193: I disagree as the fCpGs can switch back and the current data does not convince me that this is an improvement upon mosaic mutation analysis. In my mind, the main advantage is the re-analysis of existing data and the parallel functional insights that can be obtained.

Lineage analysis is more straightforward with DNA sequencing, but with an error rate of ~10-9 per base per division, one needs to sequence a billion base pairs to distinguish between immediate daughter cells. By contrast, with an inferred error rate of ~10-2 per fCpG per division, much less sequencing (about a million-fold less) is needed to find differences between daughter cells.

g) Lines 208-209: I would be careful with claims of complexity resolution given many of the limitations and inherent systematic similarities, as well as the potential of fCpGs to change back to an ancestral state later in the lineage.

Have modified the manuscript to indicate the analysis would be more challenging due to back changes.

h) There seem to be few figures that assess phenomena across the three brains. Even when they exist there is no attempt to provide any statistical analyses to support the conclusions or permutations to assess outlier status relative to expectations.

The analysis could be more extensive, but with only three brains, any results, like this study itself, would be rightly judged inadequate.

Figure 2B: there appears to be a higher number of '0s' for, for instance, inhibitory neurons compared to excitatory neurons. Is that correct and worth mentioning? The

changing axes scales also make it hard to assess.

Inhibitory neurons do appear to have more unmethylated fCpGs compared to excitatory neurons, but in general, most inhibitory fCpGs are methylated with a skew to fully methylated fCpGs, consistent with the barcode starting predominately methylated and inhibitory neurons generally appearing earlier in development relative to excitatory neurons.

j) Figure 2C: I have several issues with this. A minor one is the use of 'Glial' which, I believe, does not appear anywhere else before this, so I am unclear what this curve represents. Generally, however, I am not sure what the y-axis represents, as it is not described in the methods or figure legend. I initially thought it was the cumulative frequency, but I do not think that this squares with the data shown in B. I appreciate the overall idea of having 'earlier'/samples with fewer divisions being shifted to the left, but it is very confusing to me when I try to understand the details of the plot.

This graph is now better described in the legend. “Glial” cells are defined as oligodendrocytes and astrocytes. Other non-neuronal cells (such a microglial cells) have now been removed from the graph.

This graph attempts to illustrate how it may be possible to reconstruct brain development from adult neurons, assuming barcodes are mitotic clocks that become polymorphic with cell division. The X axis is “time”, and the Y axis indicates when different cell types reach their adult levels. The cartoon indicates what is visually present along the X axis during development--- brainstem, then ganglionic eminences with a thin cortex, and finally the mature brain with a robust cortex. Time for the X axis is barcode methylation and starts at 100% and ends at 50% or greater methylation. The fCpG barcode methylation of each cell places it on this timeline and indicates when it ceased dividing and differentiated.

The Y axis indicates the progressive accumulation of the final adult contents of each cell type during this timeline. Early in development, the brain is rudimentary and adult cells are absent. At 90% methylation, only the inhibitory neurons in the pons are present. At 80% methylation, some excitatory neurons are beginning to appear. Inhibitory neurons in the pons have reached their final adult levels and many other inhibitory neuron types are reaching adult levels. By 70% methylation, most inhibitory neurons have reached their adult levels, and more adult excitatory neurons (mainly low cortical neurons, L4-6) and glial cells are beginning to appear. By 60% methylation, inhibitory neurogenesis has largely finished. Adult excitatory neurons and glial cells are more abundant and reach their adult levels by 50% or greater cell barcode methylation levels.

The graph illustrates a rough alignment between mitotic ages inferred by barcode methylation levels and the physical appearances of different neuronal types during development. Many neurons die during development, and this graph, if valid, indicates when neurons that survive to adulthood appear during development.

k) Figure 4Bff: it is confusing to me that the text jumps to these panels after introducing Figure 5. This makes it very hard to read this section of the text.

The Figures appear in the order they are first referred to in the text.

l) Figure 5A: could any of this difference be explained by the shared lineage of excitatory neurons and dorsal neocortical glia?

Not sure

m) Figure 5B: after stating that interneurons have a higher lineage fidelity, the figure legend here states the opposite and I am somewhat confused by this statement.

The legend and text have been clarified. Fig 5A restricts fidelity to within inhibitory cell types. Fig 5B compares between neuron subtypes, and illustrates more apparent inhibitory subtype switching, albeit there are more interneuron subtypes than excitatory subtypes.

n) Figure 5E: generally, the use of tSNE for large pairwise distance analysis is often frowned upon (e.g., PMID 37590228), and I would reconsider this argument.

This analysis was an attempt to illustrate that cells of the same phenotype based on their tSNE metrics can be either closely or more distantly related. Although the tSNE comparisons were restricted to subtypes (and not to the entire tSNE graph), tSNE are not designed for such comparisons. This graph and discussion are deleted.

Reviewer #2 (Public review):

The manuscript by Shibata proposed a potentially interesting idea that variation in methylcytosine across cells can inform cellular lineage in a way similar to single nucleotide variants (SNVs). The work builds on the hypothesis that the "replication" of methylcytosine, presumably by DNMT1, is inaccurate and produces stochastic methylation variants that are inherited in a cellular lineage. Although this notion can be correct to some extent, it does not account for other mechanisms that modulate methylcytosines, such as active gain of methylation mediated by DNMT3A/B activity and activity demethylation mediated by TET activity. In some cases, it is known that the modulation of methylation is targeted by sequence-specific transcription factors. In other words, inaccurate DNMT1 activity is only one of the many potential ways that can lead to methylation variants, which fundamentally weakens the hypothesis that methylation variants can serve as a reliable lineage marker. With that being said (being skeptical of the fundamental hypothesis), I want to be as open-minded as possible and try to propose some specific analyses that might better convince me that the author is correct. However, I suspect that the concept of methylation-based lineage tracing cannot be validated without some kind of lineage tracing experiment, which has been successfully demonstrated for scRNA-seq profiling but not yet for methylation profiling (one example is Delgado et al., nature. 2022).

I thank Reviewer 2 for the careful evaluation. The validation experiment example (Delgado et al.) introduced sequence barcodes in mice, which is not generally feasible for human studies.

(1) The manuscript reported that fCpG sites are predominantly intergenic. The author should also score the overlap between fCpG sites and putative regulatory elements and report p-values. If fCpG sites commonly overlap with regulatory elements, that would increase the possibility that these sites being actively regulated by enhancer mechanisms other than maintenance methyltransferase activity.

As mentioned for Reviewer 1, fCpGs are filtered to eliminate cell type specific methylation.

(2) The overlap between fCpG and regulatory sequence is a major alternative explanation for many of the observations regarding the effectiveness of using fCpG sites to classify cell types correctly. One would expect the methylation level of thousands of enhancers to be quite effective in distinguishing cell types based on the published single-cell brain methylome works.

As mentioned above, the manuscript did not clearly indicate that the fCpG barcode is not a cell type classifier. The distinctions between fCpG barcodes and cell type classifiers are better explained in the new Supplement.

(3) The methylation level of fCpG sites is higher in hindbrain structures and lower in forebrain regions. This observation was interpreted as the hindbrain being the "root" of the methylation barcodes and, through "progressive demethylation" produced the methylation states in the forebrain. This interpretation does not match what is known about methylation dynamics in mammalian brains, in particular, there is no data supporting the process of "progressive demethylation". In fact, it is known that with the activation of DNMT3A during early postnatal development in mice or humans (Lister et al., 2013. Science), there is a global gain of methylation in both CH and CG contexts. This is part of the broader issue I see in this manuscript, which is that the model might be correct if "inaccurate mC replication" is the only force that drives methylation dynamics. But in reality, active enzymatic processes such as the activation of DNMT3A have a global impact on the methylome, and it is unclear if any signature for "inaccurate mC replication" survives the de novo methylation wave caused by DNMT3A activity.

Reviewer 2 highlights a critical potential flaw in that any ancestral signal recorded by random replication errors could be overwritten by other active methylation processes. I cannot present data that indicates fCpG replication errors are never overwritten, but new data indicate barcode reproducibility and stability with aging.

New data are also present where barcodes are compared between daughter cells (zygote to ICM) in the setting of active and passive demethylation, when germline methylation is erased. This new analysis shows that daughter cells in 2 to 8 cell embryos have more related barcodes than morula or ICM cells. The subsequent active remethylation by a wave of DNMT3A activity may underlie the observation that the barcode appears to start predominately methylated in brain progenitors.

(3) Perhaps one way the author could address comment 3 is to analyze methylome data across several developmental stages in the same brain region, to first establish that the signal of "inaccurate mC replication" is robust and does not get erased during early postnatal development when DNMT3A deposits a large amount of de novo methylation.

See above

(4) The hypothesis that methylation barcodes are homogeneous among progenitor cells and more polymorphic in derived cells is an interesting one. However, in this study, the observation was likely an artifact caused by the more granular cell types in the brain stem, intermediate granularity in inhibitory cells, and highly continuous cell types in cortical excitatory cells. So, in other words, single-cell studies typically classify hindbrain cell types that are more homogenous, and cortical excitatory cells that are much more heterogeneous. The difference in cell type granularity across brain structures is documented in several whole-brain atlas papers such as Yao et al. 2023 Nature part of the BICCN paper package.

As noted above, fCpG barcode polymorphisms and cell type differentiation are confounded because cells of the same phenotype tend to have common progenitors. The fCpG barcode is not a cell type classifier but more a cell division clock that becomes polymorphic with time. Although fCpG barcodes could be more polymorphic in cortical excitatory cells because there are many more types, fCpG barcodes would inherently become more polymorphic in excitatory cells because they appear later in development.

(5) As discussed in comment 2, the author needs to assess whether the successful classification of cell types (brain lineage) using fCpG was, in fact, driven by fCpG sites overlapping with cell-type specific regulatory elements.

Although unclear in the manuscript, the fCpG is not a cell classifier and the barcode is polymorphic between cells of the same type. fCpG barcodes can appear to be cell classifiers because cell types appear at different times during development, and therefore different cell types have characteristic average barcode methylation levels.

(6) In Figure 5E, the author tried to address the question of whether methylation barcodes inform lineage or post-mitotic methylation remodeling. The Y-axis corresponds to distances in tSNE. However, tSNE involves non-linear scaling, and the distances cannot be interpreted as biological distances. PCA distances or other types of distances computed from high-dimensional data would be more appropriate.

The Figure and discussion are deleted (similar comment by Reviewer 1)

Reviewer #3 (Public review):

Summary:

In the manuscript entitled "Human Brain Barcodes", the author sought to use single-cell CpG methylation information to trace cell lineages in the human brain.

Strengths:

Tracing cell lineages in the human brain is important but technically challenging. Lineage tracing with single-cell CpG methylation would be interesting if convincing evidence exists.

Weaknesses:

As the author noted, "DNA methylation patterns are usually copied between cell division, but the replication errors are much higher compared to base replication". This unstable nature of CpG methylation would introduce significant problems in inferring the true cell lineage. The unreliable CpG methylation status also raises the question of what the "Barcodes" refer to in the title and across this study. Barcodes should be stable in principle and not dynamic across cell generations, as defined in Reference#1. It is not convincing that the "dynamic" CpG methylation fits the "barcodes" terminology. This problem is even more concerning in the last section of results, where CpG would fluctuate in post-mitotic cells.

I thank Reviewer 3 for his thoughtful and careful evaluation. I think the "barcode" terminology is appropriate. Dynamic engineered barcodes such as CRISPR/Cas9 mutable barcodes are used in biology to record changes over time. The fCpG barcode appears to start with a single state in a progenitor cell and changes with cell division to become polymorphic in adult cells. Therefore, I think the description of a dynamic fCpG barcode is appropriate.

Reviewer #3 (Recommendations for the authors):

(1) As the author noted, "DNA methylation patterns are usually copied between cell division, but the replication errors are much higher compared to base replication". This unstable nature of CpG methylation would introduce significant problems in inferring the true cell lineage. To establish DNA methylation as a means for lineage tracing, one control experiment would be testing whether the DNA methylation patterns can faithfully

track cell lineages for in vitro differentiated & visibly tracked cell lineages. Has this kind of experiment been done in the field?

These types of experiments have not been performed to my knowledge and an appropriate tissue culture model is uncertain. New single cell WGBS data from the zygote to ICM indicate that more immediate daughter cells have more related barcodes even in the setting of active DNA demethylation.

(2) The study includes assumptions that should be backed with solid rationale, supporting evidence, or reference. Here are a couple of examples:

a) the author discarded stable CpG sites with <0.2 or >0.8 average methylation without a clear rationale in H02, and then used <0.3 and >0.7 for a specific sample H01.

The filtering was ad hoc and was used to remove, as much as possible, CpG sites with cell type specific or patient specific methylation. CpG sites with skewed methylation are more likely cell type specific, whereas X chromosome CpG sites with methylation closer to 0.5 in male cells are more likely to be unstable. The ad hoc filtering attempted to remove cell specific CpGs sites while still retaining enough CpG sites to allow comparisons between cells.

b) The author assumed that the early-formed brain stem would resemble progenitors better and have a higher average methylation level than the forebrain. However, this difference in DNA methylation status could reflect developmental timing or cell type-specific gene expression changes.

This observation that brain stem neurons that appear early in development have highly methylated fCpG barcodes in all 3 brains supports the idea that the fCpG barcode starts predominately methylated. Alternative explanations are possible.

(3) The conclusion that excitatory neurons undergo tangential migration is unclear - how far away did the author mean for the tangential direction? Lateral dispersion is known, but it would be striking that the excitatory neurons travel across different brain regions. The question is, how would the author interpret shared or divergent methylation for the same cell type across different brain regions?

As noted with Reviewer 1, this analysis is modified to indicate that evidence of tangential migration is greater for inhibitory than excitatory neurons, but the extent of excitatory neuron migration is uncertain because of sparse sampling, and because fCpG barcodes can be similar by chance.

(4) The sparsity and resolution of the single-cell DNA methylation data. The methylation status is detected in only a small fraction ($\sim 500/31,000 = 1.6\%$) of fCpGs per cell, with only 48 common sites identified between cell pairs. Given that the human genome contains over 28 million CpG sites, it is important to evaluate whether these fCpGs are truly representative. How many of these sites were considered "barcodes"?

fCpG barcodes are distinct from traditional cell type classifiers, and how fCpGs are identified are better outlined in the new Supplement.

(5) While focusing on the X-chromosome may simplify the identification of polymorphic fCpGs, the confidence in determining its methylation status (0 or 1) is questionable when a CpG site is covered by only one read. Did the author consider the read number of detected fCpGs in each cell when calculating methylation levels? Certain CpG sites on

autosomes may also have sufficient coverage and high variability across cells, meeting the selection criteria applied to X-chromosome CpGs.

In most cases, a fCpG site was covered by only a single read

(6) The overall writing in the Title, the Main text, Figure legends, and Methods sections are overly simplified, making it difficult to follow. For instance, how did the author perform PWD analysis? How did they handle missing values when constructing lineage trees?

There is not much introduction to lineage tracing in the human brain or the use of DNA methylation to trace cell lineage.

These shortcomings are improved in the manuscript and with the new Supplement. The analysis pipeline including the Python programs are outlined and included as new Supplemental materials. IQ tree can handle the binary fCpG barcode data and skips missing values with its standard settings.

Line 80: it is unclear: "Brain patterns were similar"

Clarified

Line 98: The meaning is unclear here: "Outer excitatory and glial progenitor cells are present" What are these glial progenitor cells and when/how they stop dividing?

The glial cells are the oligodendrocytes and astrocytes. The main take away point is that these glial cells have low barcode methylation, consistent with their appearances later in development.

Line 104: It is unclear if this is a conclusion or assumption -- "A progenitor cell barcode should become increasingly polymorphic with subsequent divisions." The "polymorphic" happens within the progenitors, their progenies, or their progenies at different time points.

The statement is now clarified as an assumption in the manuscript.

Similarly line 134 "Barcodes would record neuronal differentiation and migration." Is this a conclusion from this study or a citation? How is the migration part supported?

The reasoning is better explained in the manuscript. Migration can be documented if immediate daughter cells with similar barcodes are found in different parts of the adult brain, albeit analysis is confounded by sparse sampling and because barcodes may be similar by chance.

Line 148 and 150: "Nearest neighbor ... neuron pairs" in DNA methylation status would conceivably reflect their cell type-specific gene expression, how did the author distinguish this from cell lineage?

As noted above, because cells with similar phenotypes usually arise from common progenitors, cells within a clade are also usually related. However, the barcodes are still polymorphic within a clade and potentially add complementary information on mitotic ages, ancestry within a clade, and possible cell migration.

Figure 3C: "Cells that emerge early in development" Where are they on the figure?

Hindbrain neurons differentiate early in development and their barcodes are more methylated. The figure has been modified to label some of the values with their neuron types. Also, the older figure mistakenly included data from all 3 brains and now the data are only from brain H01.

| *Figures 4D and 4E, distinguishing cell subtypes is challenging, as the same color palette is used for both excitatory and inhibitory neurons.*

Unfortunate limitations due to complexity and color limitations

| *Figures 4 and 5, what are these abbreviations?*

The abbreviations are presented in Figure 1 and maintained in subsequent figures.

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